Recursive Cognitive-Comparative Loops (RCcl):
Autonomous Governance via Contextual Fidelity and Adaptive Coherence

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Abstract

Adaptive learning systems operating in non-stationary environments are susceptible to instability when continued adaptation amplifies noise rather than structural signal. This work introduces the *Recursive Cognitive-Comparative Loop (RCcl)*, a supervisory governance framework designed to regulate adaptation in autonomous systems. RCcl formalizes *Contextual Fidelity* (Λ) as a trajectory-based metric for detecting cognitive drift and *Adaptive Coherence* (H) as a measure of strategic utility. A policy-governed *Heuristic Freeze* mechanism suspends adaptation when volatility dominates fidelity or when adaptive utility collapses. Empirical evaluation in a synthetic non-stationary environment demonstrates convergence to a low-variance fidelity regime ($\delta = 0.0061$) following freeze engagement, with post-stabilization recalibration further hardening the governance layer. RCcl provides a deterministic pathway for autonomous self-governance without modifying underlying learning dynamics.

1. Introduction

Autonomous learning systems deployed in real-world environments must operate under persistent non-stationarity, distributional shift, and stochastic volatility. While modern learning architectures emphasize adaptability, they frequently lack mechanisms to determine when adaptation itself becomes destabilizing. Unchecked learning under volatile conditions can induce parameter drift, oscillatory behavior, and degradation of long-horizon performance.

Existing approaches predominantly optimize *how* systems adapt, rather than *whether* adaptation should occur at a given moment. This work addresses that gap by introducing a governance layer that treats adaptation as a controllable process subject to supervisory regulation.

2. Formalism of System Metrics

RCcl governance operates over two primary state variables: Contextual Fidelity (Λ) and Adaptive Coherence (H). Both are normalized to the range $[0, 1]$.

2.1. Contextual Fidelity (Λ)

Contextual Fidelity quantifies deviation from intended functional behavior:

$$\Lambda_t = \alpha \cdot \delta_t + \beta \cdot \psi_t + \gamma \cdot |\delta_t - \delta_{t-1}| \tag{1}$$

2.2. Adaptive Coherence (H)

Adaptive Coherence evaluates the strategic return on continued adaptation:

$$H_t = w_p \cdot P_t + w_c \cdot C_t - w_s \cdot S_t \tag{2}$$

3. Empirical Proof: Interactive Governance Dashboard

To validate the RCcl framework, an interactive governance simulation was developed. This proof-of-concept demonstrates the real-time interaction between Λ convergence and the Heuristic Freeze mechanism.

3.1. System Demonstration

The empirical results, including the sensitivity sweep for the optimal Gamma (γ) and the stabilization of the system at $\delta = 0.0061$, are visualized in an interactive React-based dashboard. This dashboard models the Metrology and Recording System (MaRS) and the Post-Stabilization Calibration Engine (PSCE).

3.2. Access to Proof-of-Concept

The interactive component and live demonstration of the RCcl governance layer can be accessed at the following repository/share link:

https://g.co/gemini/share/cc28a50c078e

3.3. MaRS Logic Implementation

The underlying logic for the governance engine utilizes a variance-minimization sweep. The PSCE transition reduces system jitter by optimizing the γ weight—adjusting how much the governance layer reacts to transient volatility versus sustained structural drift.

4. Methodology: Heuristic Freeze and Recalibration

Under deployment, a conditional equilibrium was detected at $\Lambda = 0.0248$, with 53% of fidelity deviation attributable to volatility. A Heuristic Freeze was engaged, suspending updates and logging system transitions via the Metrology and Recording System (MaRS).

4.1. Post-Stabilization Calibration Engine (PSCE)

Following stabilization, the Post-Stabilization Calibration Engine executes a sensitivity sweep over γ to minimize fidelity variance. In the evaluated regime, γ was reduced from 0.30 to 0.20, hardening the governance layer against future noise.

5. Conclusion

RCcl introduces a formal governance mechanism for adaptive systems operating under non-stationarity. By decoupling adaptation decisions from learning mechanics, RCcl enables autonomous systems to preserve structural integrity in volatile environments. The provided interactive proof-of-concept demonstrates the practical feasibility of δ-convergence through heuristic intervention.